

# LNCS Template Title

## Template number

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Cite a paper [1].

### 1 INC-CPA Game

Use `algorithm` and `algpseudocode`.

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**Algorithm 1** IND-CPA( $\mathcal{A}$ )

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```
1:  $b \leftarrow_{\$} \{0, 1\}$ 
2:  $\text{pk}, \text{sk} \leftarrow_{\$} \text{KGen}(1^\lambda)$ 
3:  $(m_0, m_1) \leftarrow \mathcal{A}(\text{pk})$ 
4:  $c \leftarrow_{\$} \text{Enc}(\text{pk}, m_b)$ 
5:  $b' \leftarrow \mathcal{A}(\text{pk}, c)$ 
6: return  $\mathbb{1}_{b=b'}$ 
```

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Or use `\procedureblock`

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IND-CPA $_{\text{Enc}}^{\mathcal{A}}(\lambda)$ 

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```
1 :  $b \leftarrow_{\$} \{0, 1\}$ 
2 :  $(\text{pk}, \text{sk}) \leftarrow_{\$} \text{KGen}(1^\lambda)$ 
3 :  $(m_0, m_1) \leftarrow_{\$} \mathcal{A}(1^\lambda, \text{pk}, c)$ 
4 :  $c \leftarrow_{\$} \text{Enc}(\text{pk}, m_b)$ 
5 :  $b' \leftarrow \mathcal{A}(1^\lambda, \text{pk}, c)$ 
6 : return  $\mathbb{1}_{b=b'}$ 
```

With a Figure environment:

$$\text{IND-CPA}_{\text{Enc}}^{\mathcal{A}}(\lambda)$$


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```

1 :  $b \leftarrow_{\$} \{0, 1\}$ 
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5 :  $b' \leftarrow \mathcal{A}(1^\lambda, \text{pk}, c)$ 
6 : return  $\mathbb{1}_{b=b'}$ 

```

**Fig. 1.** IND-CPAGame

## References

1. Chen, K.Y., Chen, J.P.: IACR Transactions on Cryptographic Hardware and Embedded Systems **2024**(2), 276–303 (Mar 2024). <https://doi.org/10.46586/tches.v2024.i2.276-303>, <https://tches.iacr.org/index.php/TCHES/article/view/11428>